

K Most Frequent Strings

Find the k most frequently occurring strings in an array, and return them sorted by frequency in descending order. If two strings have the same frequency, sort them in lexicographical order.

Example:

```
Input: strs = ['go', 'coding', 'byte', 'byte', 'go', 'interview', 'go'], k = 2
Output: ['go', 'byte']
```

Explanation: The strings "go" and "byte" appear the most frequently, with frequencies of 3 and 2, respectively.

Constraints:

- $k \leq n$, where n denotes the length of the array.

Intuition - Max-Heap

The two main challenges to this problem are:

- Identifying the k most frequent strings.
- Sorting those strings first by frequency and then lexicographically.

For now, let's concentrate on identifying the most frequent strings and address lexicographical ordering afterward.

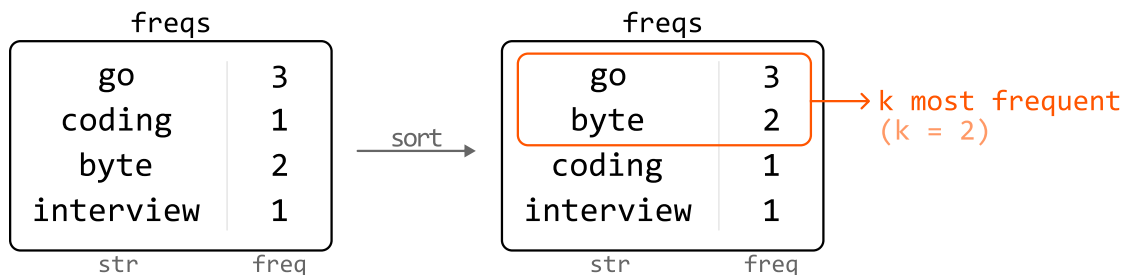
First, we need a way to keep track of the frequencies of each string. We can use a hash map for this, where the keys represent the strings and the values represent frequencies:

	freqs
go	3
coding	1
byte	2

[go coding byte byte go interview go] →

byte	2
interview	1
str	freq

The most straightforward approach is to obtain an array containing the strings from the hash map, sorted by frequency in descending order. The k most frequent strings would be the first k strings in this array.



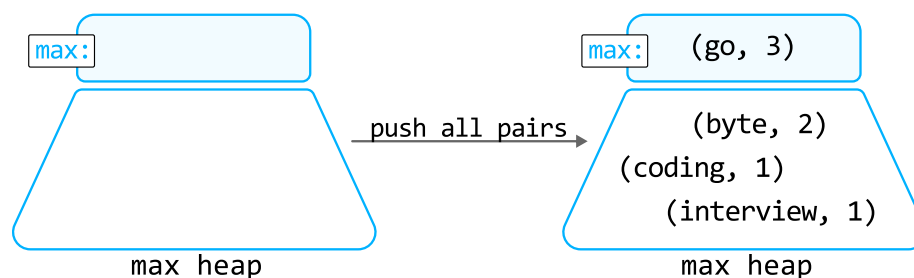
The main inefficiency of this solution is that it involves sorting all n strings, even though we only need the top k frequent ones to be sorted.

Something useful to consider: if we remove the most frequent string, the new most frequent string after this removal represents the second-most frequent overall. By repeatedly identifying and removing the most frequent string k times, we efficiently obtain our answer.

To implement this idea, we need a data structure that allows efficient access to the most frequent string at any time. A **max-heap** is perfect for this.

Max-heap

Let's find the k most frequent strings from the previous input, this time using a max-heap. First, populate the heap with each string along with their frequencies.

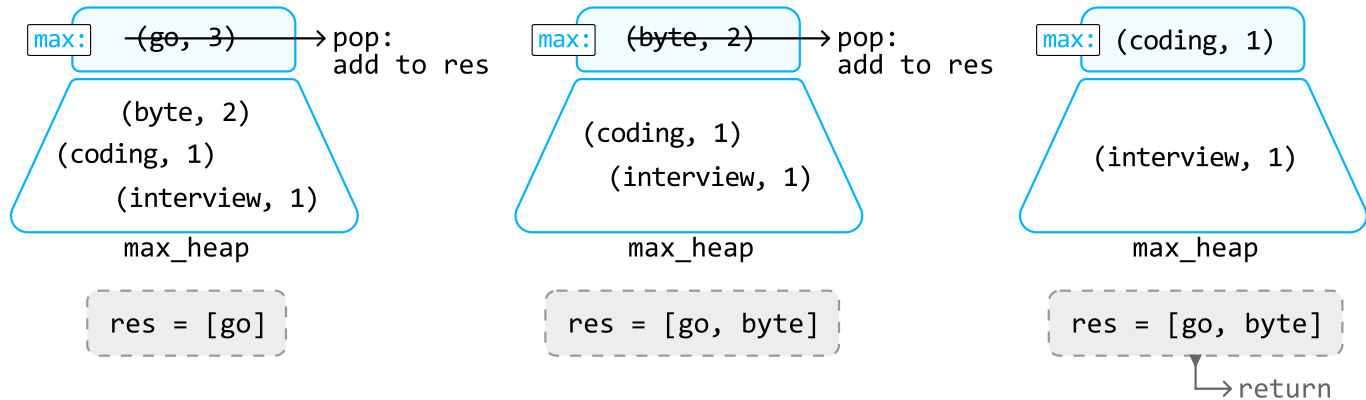


One way to populate the heap is to push all n strings into it one by one, which will take $O(n \log(n))$ time. Instead, we can perform the **heapify** operation on an array containing all the string-frequency pairs to create the max-heap in $O(n)$ time.

To collect the k most frequent strings, pop off the top element from the heap k times and store the

corresponding strings in the output array **res**:

k = 2



Now, we just need to ensure that when two strings have the same frequency, the one that comes first lexicographically has a higher priority in the heap. To do this, we can define a custom comparator for the heap that prioritizes strings lexicographically when their frequencies match, as demonstrated in the implementation below.

Implementation - Max-Heap

We create a **Pair** class for string-frequency pairs, enabling us to customize priority using a custom comparator.

Python

Java

```
from typing import List
from collections import Counter
import heapq

class Pair:
    def __init__(self, str, freq):
        self.str = str
        self.freq = freq

    # Define a custom comparator.
    def __lt__(self, other):
        # Prioritize lexicographical order for strings with equal frequencies.
        if self.freq == other.freq:
            return self.str < other.str
        # Otherwise, prioritize strings with higher frequencies.
        return self.freq > other.freq
```

```
def k_most_frequent_strings_max_heap(strs: List[str], k: int) → List[str]:
    # We use 'Counter' to create a hash map that counts the frequency of each string
    freqs = Counter(strs)
    # Create the max heap by performing heapify on all string-frequency pairs.
    max_heap = [Pair(str, freq) for str, freq in freqs.items()]
    heapq.heapify(max_heap)
    # Pop the most frequent string off the heap 'k' times and return these 'k' most
    # frequent strings.
    return [heapq.heappop(max_heap).str for _ in range(k)]
```

Complexity Analysis

Time complexity: The time complexity of `k_most_frequent_strings_max_heap` is $O(n + k \log(n))$.

- It takes $O(n)$ time to count the frequency of each string using `Counter`, and to build the `max_heap`.
- We also pop off the top of the heap k times, with each `pop` operation taking $O(\log(n))$ time.

Therefore, the overall time complexity is $O(n) + kO(\log(n)) = O(n + k \log(n))$.

Space complexity: The space complexity is $O(n)$ because the hash map and heap store at most n pairs. Note that the output array is not considered in the space complexity.

Intuition - Min-Heap

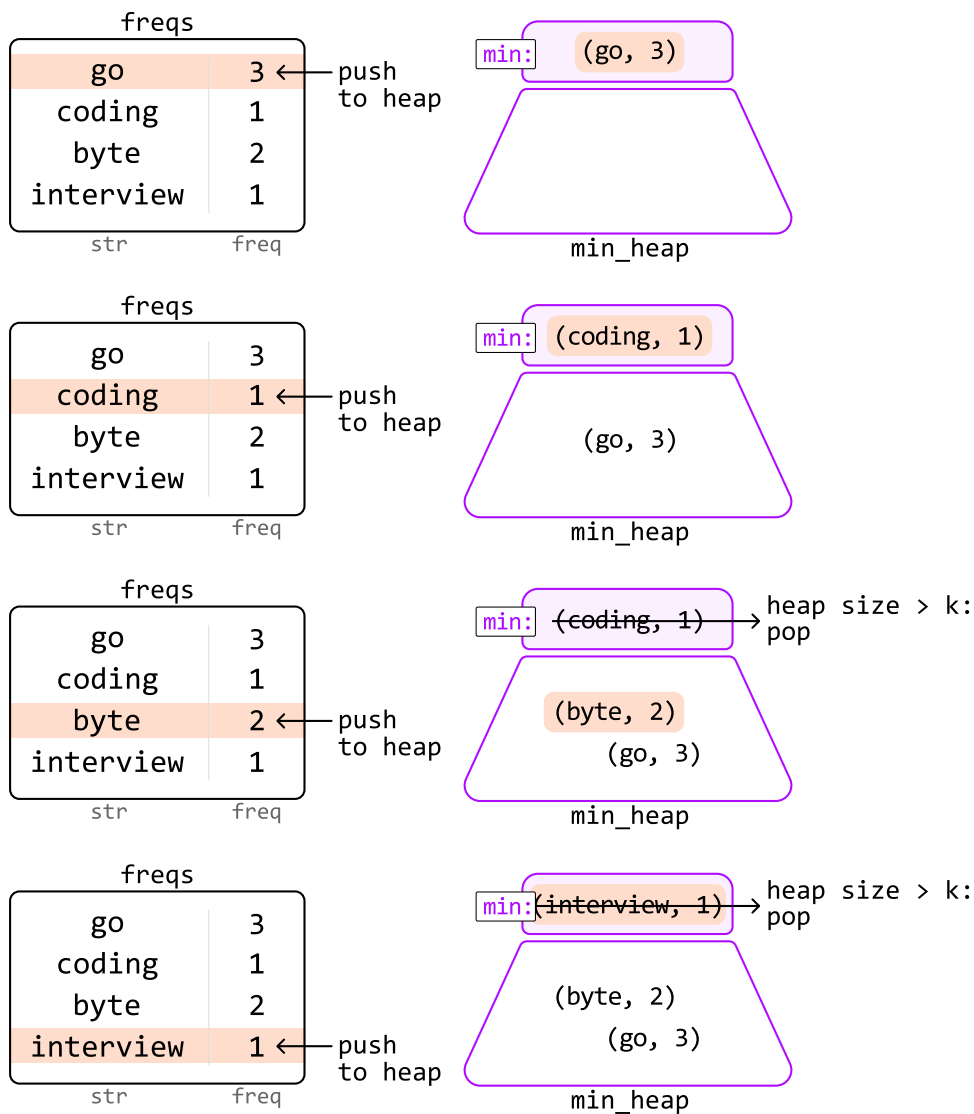
As a follow up, your interviewer may ask you to modify your solution to reduce the space used by the heap.

In the previous approach, we ended up storing up to n items in the heap. However, since we only need the k most frequent characters, is there a way to maintain a heap with a space complexity of $O(k)$?

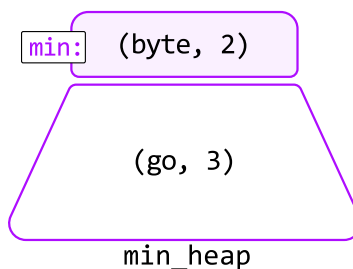
An important observation is that when our heap exceeds size k , we can **discard the lowest frequency strings until the heap's size is reduced to k again**. We can do this because those discarded strings definitely won't be among the k most frequent strings.

However, we can't implement this strategy with a max-heap because we won't have access to the lowest frequency string. Instead, we need to use a **min-heap**.

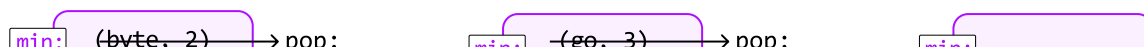
Let's observe how this works over an example:



In the end, the strings remaining in the heap are our top k frequent strings:



To retrieve these strings, pop them from the heap until it's empty. Because we're using a min-heap, we're popping off the less frequent strings first. So, we need to reverse the order of the retrieved strings before returning the result:





Python

Java

```

from typing import List
from collections import Counter
import heapq

class Pair:
    def __init__(self, str, freq):
        self.str = str
        self.freq = freq

    # Since this is a min-heap comparator, we can use the same comparator as the
    # used in the max-heap, but reversing the inequality signs to invert the priority.
    def __lt__(self, other):
        if self.freq == other.freq:
            return self.str > other.str
        return self.freq < other.freq

def k_most_frequent_strings_min_heap(strs: List[str], k: int) → List[str]:
    freqs = Counter(strs)
    min_heap = []
    for str, freq in freqs.items():
        heapq.heappush(min_heap, Pair(str, freq))
        # If heap size exceeds 'k', pop the lowest frequency string to ensure the
        # only contains the 'k' most frequent words so far.
        if len(min_heap) > k:
            heapq.heappop(min_heap)

    # Return the 'k' most frequent strings by popping the remaining 'k' strings from
    # the heap. Since we're using a min-heap, we need to reverse the result after
    # popping the elements to ensure the most frequent strings are listed first.
    res = [heapq.heappop(min_heap).str for _ in range(k)]
    res.reverse()
    return res

```

Complexity Analysis

Time complexity: The time complexity of `k_most_frequent_strings_min_heap` is $O(n \log(k))$.

- It takes $O(n)$ time to count the frequency of each string using `Counter`.
- To populate the heap, we push n words onto it, with each `push` and `pop` operation taking $O(\log(k))$ time. This takes $O(n \log(k))$ time.
- Then, we extract k strings from the heap by performing the `pop` operation k times. This takes $O(k \log(k))$ time.
- Finally, we reverse the output array, which takes $O(k)$ time.

Therefore, the overall time complexity is $O(n) + O(n \log(k)) + O(k \log(k)) + O(k) = O(n \log(k))$.

Space complexity: The space complexity is $O(n)$ because the hash map stores at most n pairs, whereas the heap only takes up $O(k)$ space. The `res` array is not considered in the space complexity.